

MUSP-BENCH: AN ADVANCED MULTIMODAL MUSIC SCORE-PERFORMANCE UNDERSTANDING BENCHMARK

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ABSTRACT

Musicians commonly communicate music through two complementary modalities: scores and performances. Scores encode musical structure and intent, while performances realize them in sound. To investigate whether models can meaningfully engage with both modalities, we introduce **MuSP-Bench**, a human-authored benchmark of 490 questions targeting understanding across **Musical Scores** and **Performances**. The benchmark provides a comprehensive view of how models understand and relate these modalities. It distinguishes itself from existing benchmarks through its long-horizon understanding tasks, real-world inquiries across modalities, and musical interpretation questions across classical piano and orchestral works. We evaluate frontier multimodal large language models under multiple input conditions and formats. Our results show that these models struggle substantially to understand scores, while facing even greater challenges when reasoning about performance audio. The benchmark is available at ...

1. INTRODUCTION

Musicians create and communicate musical ideas through both scores and performances. Scores encode compositional intent—pitch, rhythm, harmony, dynamics, articulation, and structure—while performance realizes them through timing, phrasing, voicing, technique, and timbre. Musicians continually move between these representations: reading, imagining, performing, listening, and returning to the score. Models aiming at human-level music understanding and serving real-world music applications, including education and artistic practice should therefore be able to reason across both score and audio.

Existing benchmarks largely assess these modalities in isolation. MSU-Bench [1] evaluates ABC and image-based scores, but mostly through short-horizon or true/false questions. MusicTheoryBench [2], Ziqi-Eval [3], and WildScore [4] similarly focus on ABC or single-image excerpts and predominantly use multiple-choice

evaluation. MuxQA [5] and SSMR-Bench [6] generate synthetic scores and questions on properties such as pitch, rhythm, chords, scales, and meter. While useful for symbolic understanding, these benchmarks provide limited coverage of higher-level musical interpretation. MuseBench [7] is one of the few benchmarks to jointly evaluate score and performance. However, its joint tasks mainly test whether tempo matches a reference, notes are omitted, or performance speed is consistent through true/false questions. As such, it does not address dimensions such as phrasing, voicing, rubato, technique, and expression. Benchmarks like HumMusQa [8], MUSE [9], CMI-Bench [10] test musical perception from audio, including rhythm, melody, harmony, tempo, and instrumentation. However, these benchmarks rarely assess the models’ understanding of the performance: how phrasing, dynamics, or expressive character shape a particular realization of a work.

Overall, existing benchmarks rarely operate at the level of complete musical works or the questions musicians ask of them: how phrases unfold, where interpretive and technical challenges arise, and how a performance relates to the written score. We therefore introduce a multimodal benchmark that combines local musical understanding with higher-level reasoning and real-world inquiries across complete works and across score and performance.

2. BENCHMARK

MuSP-Bench comprises 490 questions across 24 classical piano and orchestral works. Professional musicians authored questions requiring information from the score, performance, or both. Each specifies an answer format, such as a pitch, chord progression, timestamp, measure range, or short phrase, and may allow multiple correct answers to reflect musical ambiguity. Piano scores come from ASAP [11], corrected using aligned PianoCoRe MIDI [12]; orchestral scores and audio come from BSED [13]. Further details are available on the demo website.

Each question is annotated along four dimensions. **Modality** indicates whether the minimal required information comes from the score (S), performance (P), both (SP), or either (S/P). **Horizon** captures the smallest span over which evidence must be integrated: short (1–2 measures or less than 5 s), medium (3–16 measures or between 5–30 s), long (larger, noncontiguous, or piece-level), or any. Pieces average 10 score pages (1–40) and nearly 5



minutes (0:17–12:29). **Content** follows a hierarchy from *pitch*, through *temporal organization*, *orchestration*, and *performance realization*, to *melodic*, *harmonic*, *formal*, *interpretive*, and *contextual* understanding. Finally, **Action** specifies the reasoning operation: identification, localization, quantification, comparison, inference, transcription, or explanation/summarization.

2.1 Question Overview

Representative score questions (S) assess harmonic functions, chord progressions, motifs, phrasing, interpretation, and voicing. Performance questions assess understanding of the played music (P or S/P) and its realization of the score (SP), including rubato, expressive emphasis, tone color, melodic variation, phrasing, voicing, fingering, and pedaling. Representative questions ask which note receives the strongest expressive emphasis, where a melody is varied, how a performer shapes a phrase, or which pitches constitute a secondary voice. MuSP-Bench also tests musical knowledge by asking models to identify works from musical content alone, without metadata, from either score or audio (S/P).

Examples drawn from SP modality:

- Consider the theme starting at 2:51. In which bar does the performer reach the subverted, silent climax of this theme, characterized by a significant slowdown in tempo and loudness?
- In bars 224–226, one melodic note stands out expressively by introducing a substantially warmer tone color than the preceding fragment. At what timestamp does this change in color occur?

3. EXPERIMENTS

We evaluate multimodal foundation models using the input modalities they support: GPT 5.6 Sol and Qwen 3.6 Plus on text and score images, Audio Flamingo Next on audio, and Muse Spark 1.2 and Qwen-Omni-Plus on text, image, and performance audio. Input depends on the modality. For score-only questions, we use ABC notation and PDF-scores converted to images; for performance-only questions, audio and MIDI rendered as text. For score-performance questions, we combine score images with audio and ABC notation with audio. A language-only baseline measures how far models can answer from prior knowledge of a work without receiving the score or performance, referred to as ‘context only’ in Table 1.

4. RESULTS

On score-only modality, GPT-5.6-Sol performs best overall (50.6% on ABC and 26.1% on image), whereas the remaining models achieve substantially lower scores. Processing of scores did not have a significant impact on score-reading ability. Scores on the performance-only tier varies strongly by representation: Qwen3.5-Omni Plus obtains 11.3% on audio, whereas GPT achieves 50% using

MIDI represented in text format, affirming that MIDI approximates the performance to some extent. Involving audio in SP questions usually dramatically degrades performance compared to MIDI. For the S/P modality, Qwen Omni achieves 50% accuracy using audio, higher than score, whereas GPT outperforms with scores. This is

Table 1. Accuracy (%) by subset and input condition. The #Q column gives the number of eligible questions. Bold marks the highest result in each row; underlining marks each model’s highest input result within that subset. N/A denotes an unsupported input format or task. The original S/P questions are separated into general S/P, Tonality/style, and Composition subsets. Composition contains only Composer and Work questions. The final aggregates exclude Context-only baselines and weight the six subsets equally.

Subset	Input	#Q	GPT-5.6 Sol	Qwen3.5 Omni Plus	MuseSpark 1.2	Qwen-3.6 Plus	Audio Flamingo Next
Score only (S)	Context only	180	5.0	1.7	4.4	1.7	N/A
	ABC	180	50.6	11.7	<u>31.7</u>	11.1	N/A
	Image	180	26.1	<u>12.2</u>	15.0	<u>13.3</u>	N/A
Performance only (P)	Context only	106	5.7	3.8	5.7	3.8	2.8
	MIDI-as-text	106	<u>50.0</u>	10.4	<u>26.4</u>	<u>10.4</u>	N/A
	Audio	106	N/A	11.3	3.8	N/A	<u>0.9</u>
Score and performance (SP)	Context only	72	6.9	2.8	4.2	1.4	N/A
	ABC+MIDI	72	<u>43.1</u>	5.6	<u>26.4</u>	6.9	N/A
	Image+audio	72	N/A	5.6	6.9	N/A	N/A
Either (S/P)	ABC+audio	72	N/A	4.2	6.9	N/A	N/A
	Context only	36	13.9	5.6	8.3	5.6	2.8
	ABC	36	44.4	<u>16.7</u>	33.3	11.1	N/A
	MIDI-as-text	36	50.0	13.9	<u>36.1</u>	11.1	N/A
	Image	36	19.4	<u>16.7</u>	11.1	19.4	N/A
Tonality/style	Audio	36	N/A	16.7	11.1	N/A	<u>0.0</u>
	ABC	48	77.1	50.0	68.8	<u>50.0</u>	N/A
	MIDI-as-text	48	72.9	39.6	54.2	33.3	N/A
	Image	48	68.8	50.0	85.4	47.9	N/A
Context/composition	Audio	48	N/A	75.0	20.8	N/A	<u>22.9</u>
	ABC	48	35.4	12.5	43.8	8.3	N/A
	MIDI-as-text	48	27.1	0.0	27.1	0.0	N/A
	Image	48	54.2	14.6	<u>52.1</u>	<u>16.7</u>	N/A
Aggregate	Audio	48	N/A	39.6	2.1	N/A	<u>0.0</u>
	Context only	394	6.3	2.8	5.1	2.5	2.8/142
	ABC	312	51.6	18.3	39.4	16.7	N/A
	MIDI-as-text	238	50.0	14.7	33.6	13.0	N/A
	Image	312	36.2	18.9	31.1	19.9	N/A
	Audio	238	N/A	30.7	8.0	N/A	5.0
	ABC+MIDI	72	43.1	5.6	26.4	6.9	N/A
Aggregate	Image+audio	72	N/A	5.6	6.9	N/A	N/A
	ABC+audio	72	N/A	4.2	6.9	N/A	N/A
Aggregate	Available-run mean	6 subsets	46.9	19.0	27.2	15.9	N/A
	Top score per subset	6 subsets	54.2	26.7	43.0	19.5	N/A

5. CONCLUSION

We introduced MuSP-Bench, a human-authored benchmark for analytical and interpretive reasoning across musical scores and performances. Our results show that current models struggle with image-based scores and expressive performance audio. MuSP-Bench provides a foundation for measuring progress toward more meaningful musical understanding across modalities.

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